HIGH Priority (Must Have - Core Gameplay)

UC-1: Read Driver Details

ID: UC-1

Title: Read Driver Details

Description: Returns a driver's profile and attributes for display in driver/team screens.

Actors:

- User (player or system client)

Triggers:

- User opens a driver profile screen or requests driver details via API.

Pre-conditions:

- Driver exists in the database.

Post-Conditions:

- Driver details are returned and displayed to the user.

Normal Flow:

1. User requests driver details by driver ID.
2. System fetches driver record and current contract/status.
3. System returns driver attributes, history, and availability.

Alternative Flows:

- Driver not found: system returns 404.

Exceptions:

- Database unavailable: system returns 503 and logs error.

Includes: None.

Extends: None.

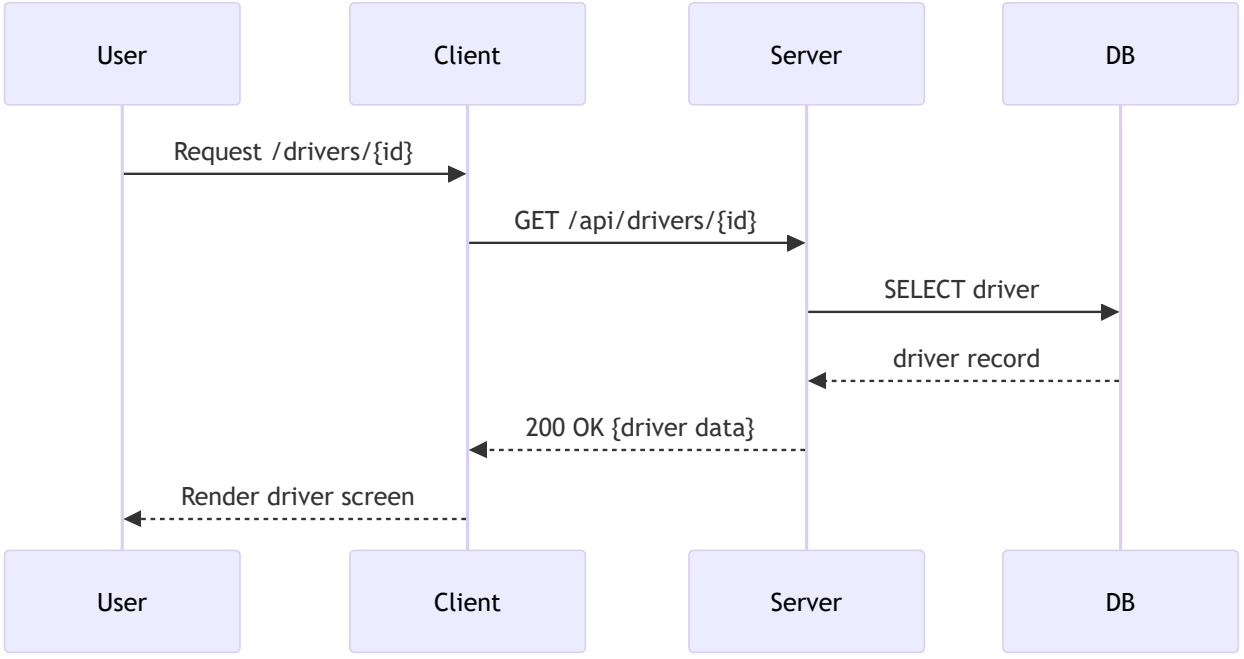
Special Conditions:

- PII rules apply to personal fields.

Acceptance Criteria:

- Valid driver ID returns complete profile JSON within acceptable latency.

Mermaid Diagram:



UC-3: Read Race Configuration

ID: UC-3

Title: Read Race Configuration

Description: Retrieve race configuration (track, weather, lap count, rules) used for simulation and UI.

Actors:

- User or Simulation Service

Triggers:

- Client requests race configuration before a simulation or for display.

Pre-conditions:

- Race configuration exists for the requested event.

Post-Conditions:

- Configuration is returned to the caller.

Normal Flow:

1. Client requests configuration for event ID.
2. Server validates access and fetches race config.
3. Server returns configuration JSON.

Alternative Flows:

- Config missing: returns 404.

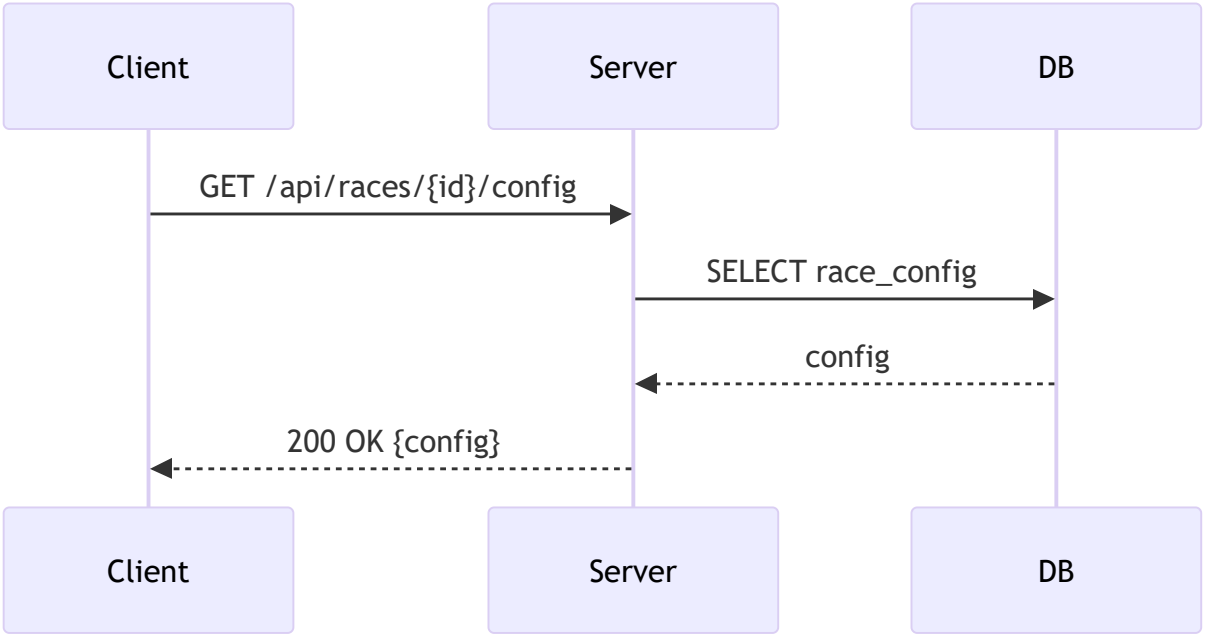
Exceptions:

- DB error: 503 returned.

Acceptance Criteria:

- Race config JSON contains track id, laps, weather, ruleset.

Mermaid Diagram:



UC-4: Create Race Result

ID: UC-4

Title: Create Race Result

Description: Store the outcome of a completed race in the database (positions, times, incidents).

Actors:

- Simulation Service (or UI submitting manual result)

Triggers:

- A race simulation finishes or manual result entered.

Pre-conditions:

- Race event exists and is in a state allowing results to be recorded.

Post-Conditions:

- Race result record created; related championship standings updated (UC-7) optionally queued.

Normal Flow:

1. Simulation posts race result payload to server.
2. Server validates schema and event state.
3. Server stores results and marks event complete.
4. Server enqueues/updates championship standing calculations.
5. Server returns acknowledgment.

Alternative Flows:

- Validation fails: return 400 with errors.
- Event already finalised: return 409 conflict.

Exceptions:

- DB write error: 500 and transaction rolled back.

Includes:

- Notification to interested clients (websocket/event bus).

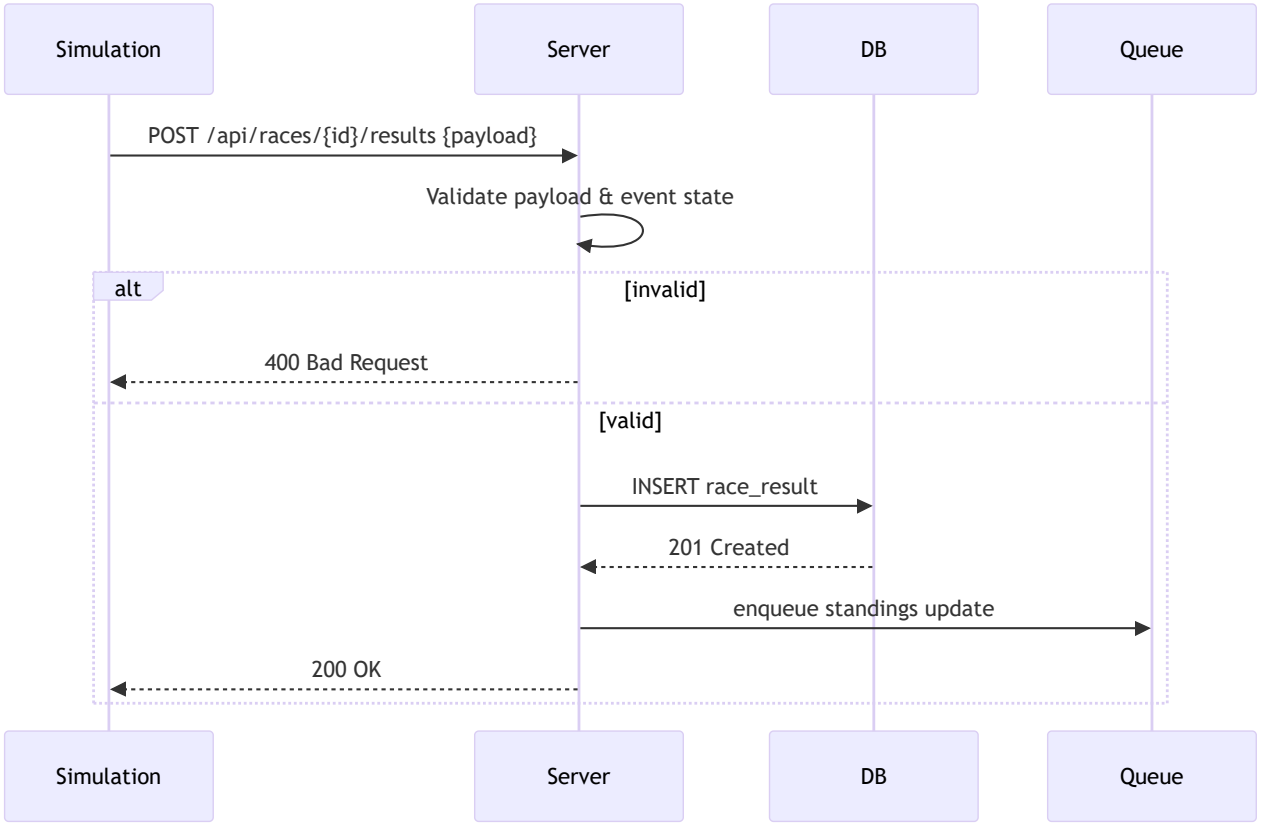
Extends:

- Trigger UC-7 to update standings.

Acceptance Criteria:

- Valid results persist and event marked completed; duplicate submissions with identical signature are ignored.

Mermaid Diagram:



MEDIUM Priority (Career Mode Depth)

UC-5: Read Race Results

ID: UC-5

Title: Read Race Results

Description: Return stored race results for a completed event for display and historical records.

Actors:

- User, Standings Service

Triggers:

- User views previous event results or an API consumer requests results.

Pre-conditions:

- Results for the event exist.

Post-Conditions:

- Results delivered to the requester.

Normal Flow:

1. Client requests results by event ID.
2. Server retrieves results and related data (lap times, penalties).
3. Server returns results.

Alternative Flows:

- No results: 404.

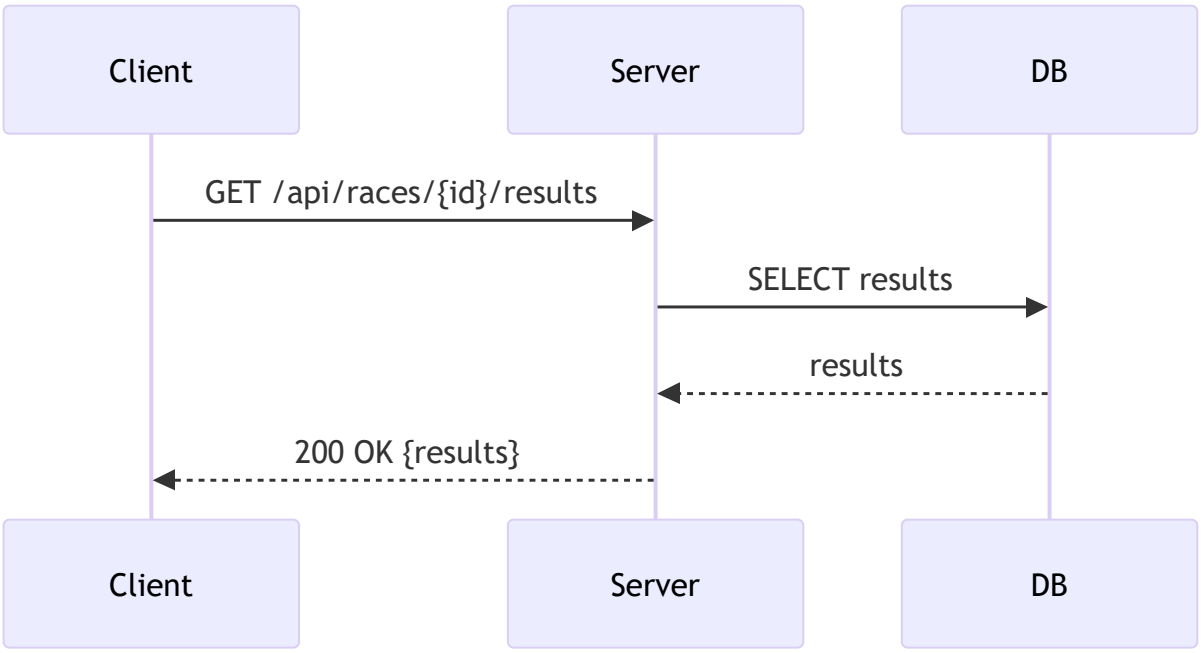
Exceptions:

- DB read error: 503.

Acceptance Criteria:

- Results endpoint returns full classification for event.

Mermaid Diagram:



UC-6: Read Championship Details

ID: UC-6

Title: Read Championship Details

Description: Return championship metadata, calendar, teams, and standings snapshot.

Actors:

- User, UI, External Client

Triggers:

- User opens championship screen.

Pre-conditions:

- Championship exists.

Post-Conditions:

- Details delivered to client.

Normal Flow:

1. Client requests championship by id.
2. Server aggregates metadata, calendar, and current standings.

3. Server returns the combined payload.

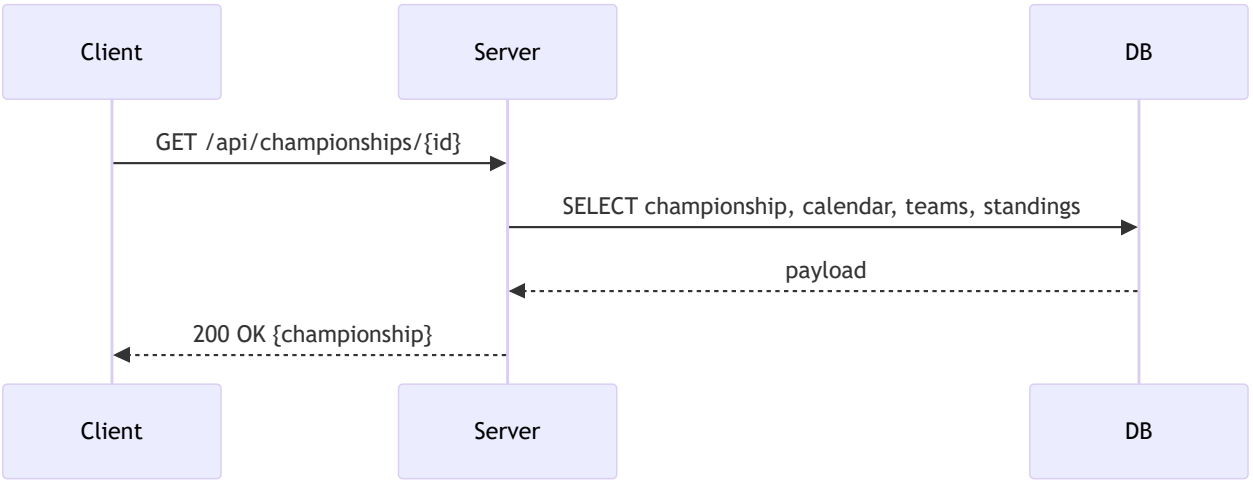
Alternative Flows:

- Championship archived: returns read-only flag.

Acceptance Criteria:

- Payload includes name, season, calendar, participating teams and current standings.

Mermaid Diagram:



UC-7: Update Championship Standings

ID: UC-7

Title: Update Championship Standings

Description: Recalculate and persist championship standings after results are created or adjusted.

Actors:

- Standings Service / Server

Triggers:

- Race result inserted/changed (UC-4) or manual correction.

Pre-conditions:

- Championship and related results are present.

Post-Conditions:

- Standings updated and stored; notifications sent.

Normal Flow:

1. Event triggers standings recalculation job.
2. Service gathers results and applies points rules.
3. Service persists standings and publishes update event.

Alternative Flows:

- Partial update: only affected rounds recomputed.

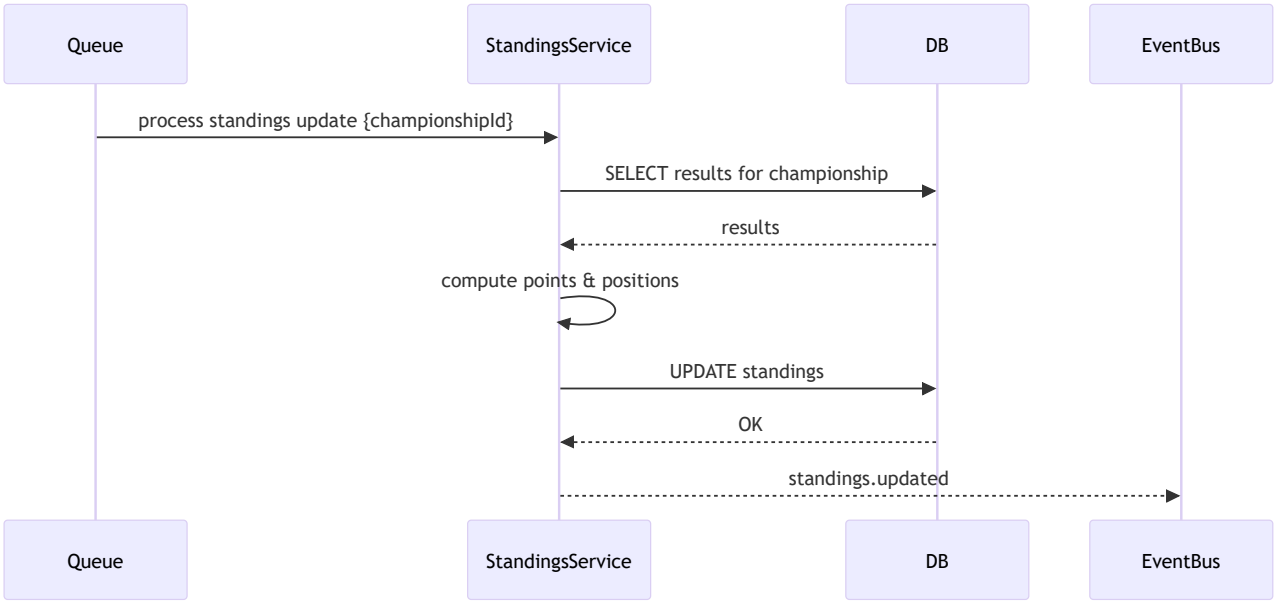
Exceptions:

- Points rule missing: log and abort update.

Acceptance Criteria:

- Standings reflect latest results and are consistent with rules.

Mermaid Diagram:



UC-8: Create Race Event

ID: UC-8

Title: Create Race Event

Description: Add a race event to a championship calendar (track, date, configuration).

Actors:

- Game Manager (user/admin) or automated scheduler

Triggers:

- User creates a new event via UI or API.

Pre-conditions:

- Championship exists and is editable.

Post-Conditions:

- Event added to calendar; notifications to teams if applicable.

Normal Flow:

1. Admin posts new event details.
2. Server validates dates and track availability.
3. Server inserts event and returns created resource.

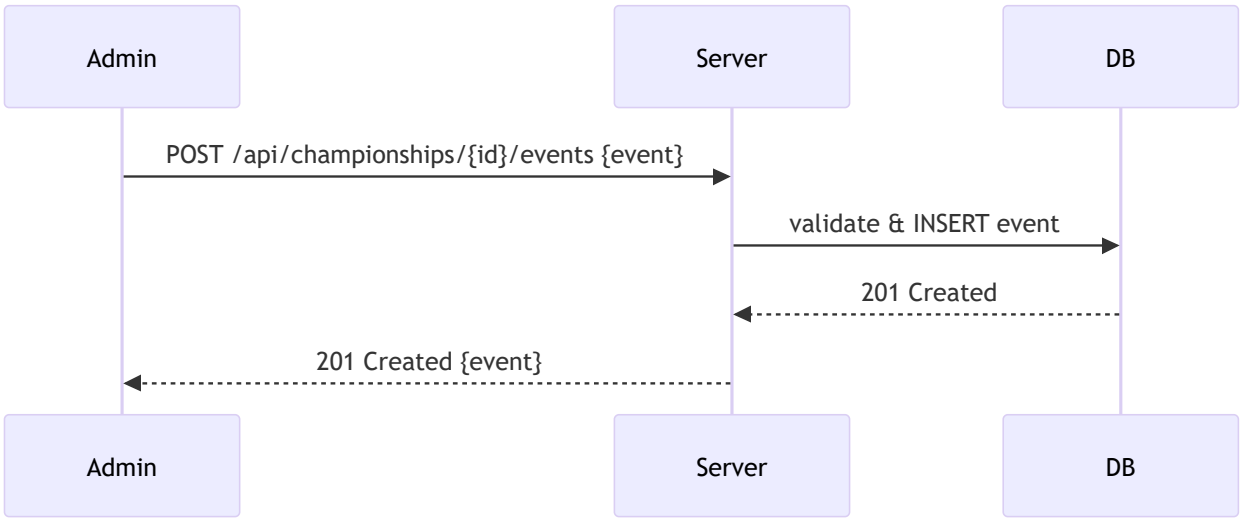
Alternative Flows:

- Date conflict: return 409 with suggested alternatives.

Acceptance Criteria:

- Calendar shows new event and clients can fetch it.

Mermaid Diagram:



UC-9: Read Race Schedule

ID: UC-9

Title: Read Race Schedule

Description: Retrieve championship calendar/schedule for UI display.

Actors:

- User, UI

Triggers:

- User opens calendar screen.

Pre-conditions:

- Championship or season exists.

Post-Conditions:

- Schedule delivered to the requester.

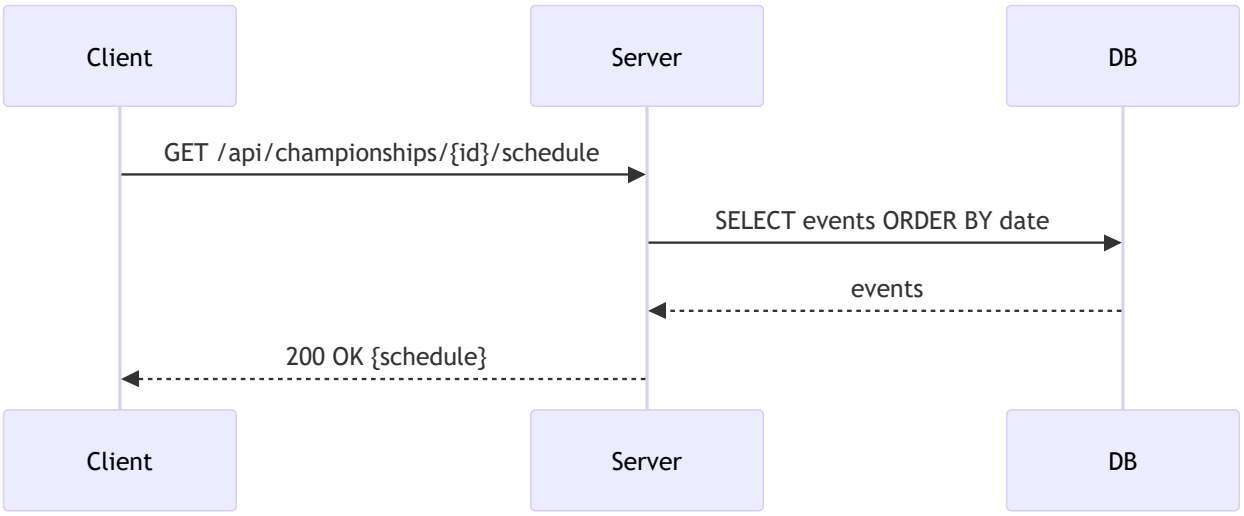
Normal Flow:

1. Client requests schedule for championship or season.
2. Server returns ordered list of events with metadata.

Acceptance Criteria:

- Schedule includes dates, track names, and event status.

Mermaid Diagram:



UC-10: Update Race Status

ID: UC-10

Title: Update Race Status

Description: Change a race event status (scheduled, in-progress, completed, cancelled).

Actors:

- Simulation Service, Admin, Scheduler

Triggers:

- Race lifecycle events or admin action.

Pre-conditions:

- Event exists and caller has permissions.

Post-Conditions:

- Event status updated; dependent systems notified.

Normal Flow:

1. Service posts status update.
2. Server validates transition and updates event record.

3. Server emits event bus notification.

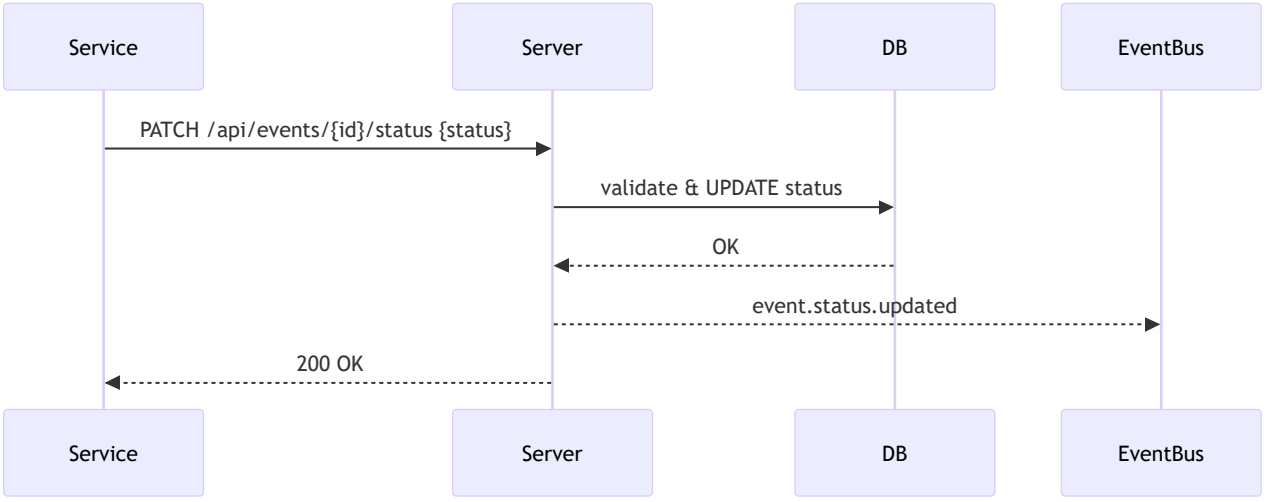
Alternative Flows:

- Invalid transition: return 400.

Acceptance Criteria:

- Status transitions follow defined lifecycle and notifications delivered.

Mermaid Diagram:



UC-11: Create Championship

ID: UC-11

Title: Create Championship

Description: Create a new championship/season with base rules and empty calendar.

Actors:

- Admin or Game Creator

Triggers:

- Admin submits new championship request.

Pre-conditions:

- Admin authenticated and allowed to create championships.

Post-Conditions:

- Championship record created and available for event creation.

Normal Flow:

1. Admin posts championship definition (name, rules, season year).
2. Server validates rules and creates resource.
3. Server returns created championship.

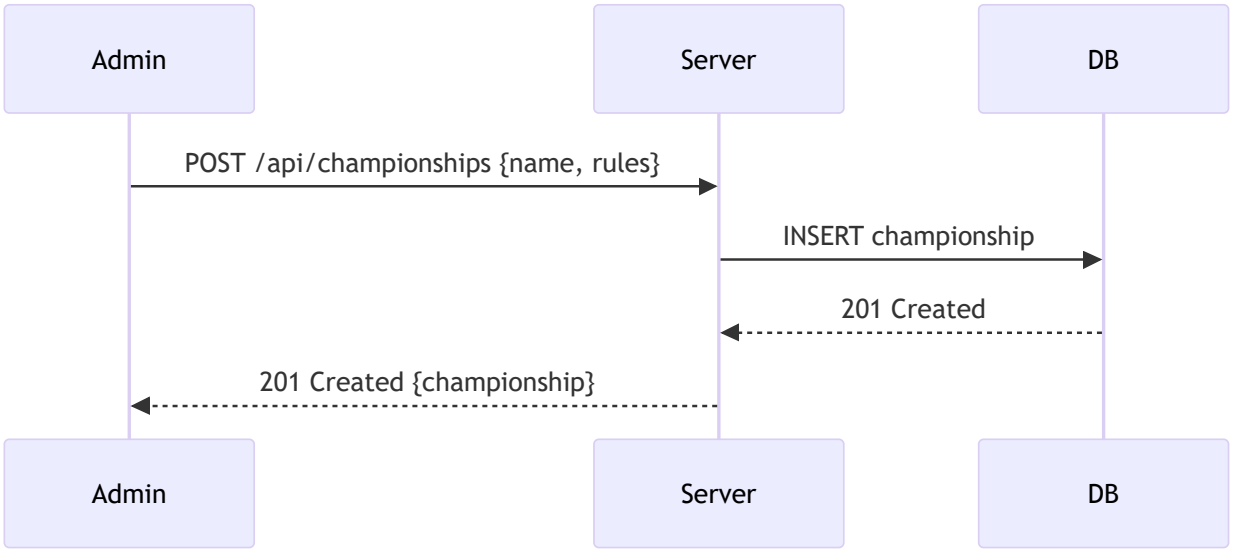
Alternative Flows:

- Invalid rules: return 400 with errors.

Acceptance Criteria:

- New championship appears in list and is editable.

Mermaid Diagram:



UC-2: Read Team Details

ID: UC-2

Title: Read Team Details

Description: Return full team profile for team screen (budget, staff, vehicles, drivers).

Actors:

- User, UI

Triggers:

- User opens team screen.

Pre-conditions:

- Team exists.

Post-Conditions:

- Team details shown in UI.

Normal Flow:

1. Client requests team by ID.
2. Server fetches team, staff, budget, roster.
3. Server returns aggregated payload.

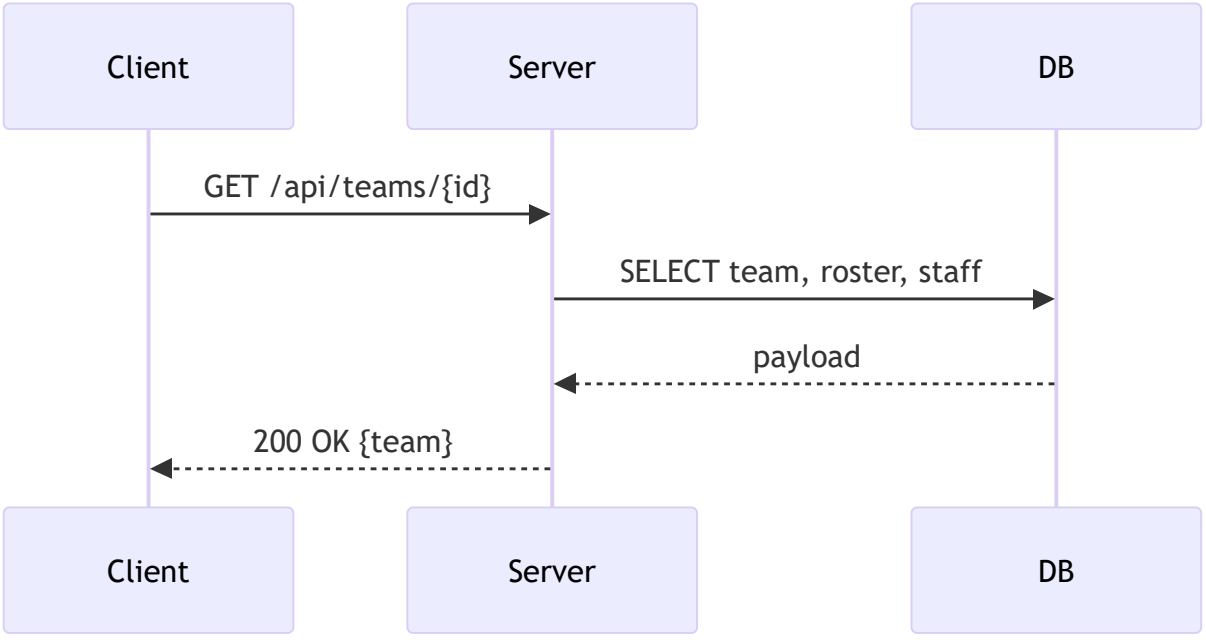
Alternative Flows:

- Team private or restricted: return 403.

Acceptance Criteria:

- Payload contains roster and core team attributes.

Mermaid Diagram:



UC-12: Create New Team

ID: UC-12

Title: Create New Team

Description: Create a new team record (use existing driver/staff pools or generate defaults).

Actors:

- User/Admin

Triggers:

- Admin or player creates a team.

Pre-conditions:

- Permissions granted; required seed data available.

Post-Conditions:

- Team created and assigned initial assets.

Normal Flow:

1. Client posts new team data.
2. Server validates and reserves required resources (budget, HQ).
3. Server inserts team and returns created entity.

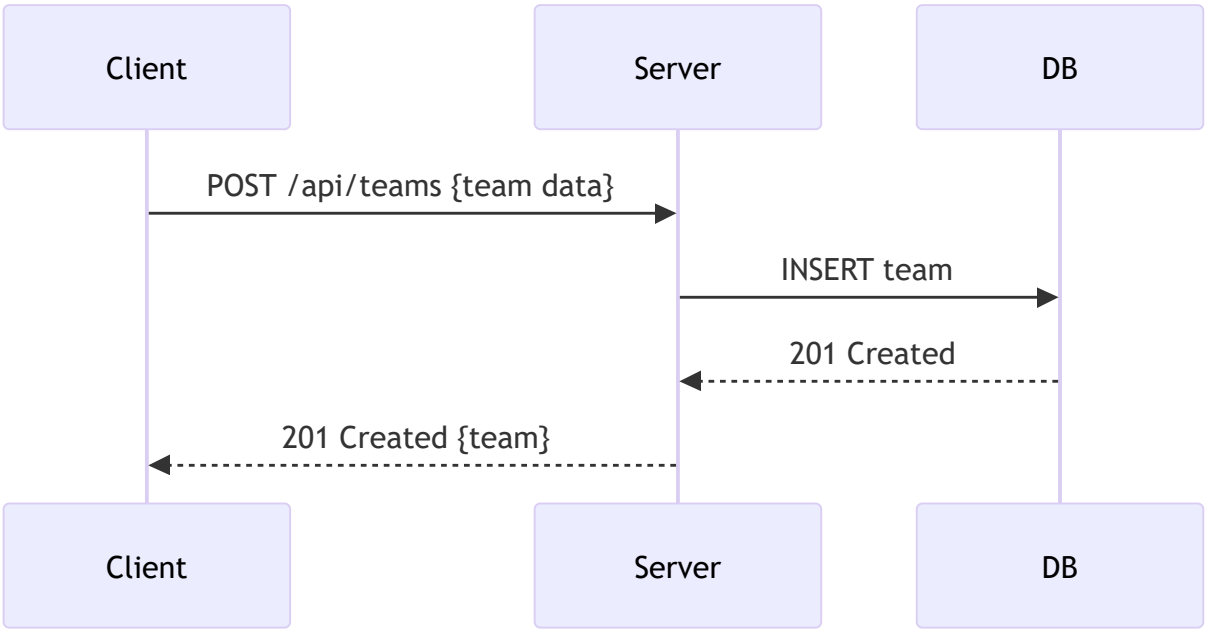
Alternative Flows:

- Resource conflict: suggest alternatives.

Acceptance Criteria:

- Team created with roster or placeholders and visible in team list.

Mermaid Diagram:



UC-13: Read Team Details (team screen)

ID: UC-13

Title: Read Team Details (team screen)

Description: Similar to UC-2 but focused on the player's own team with editable fields and affordances.

Actors:

- Player

Triggers:

- Player opens their team management screen.

Pre-conditions:

- Player has a team.

Post-Conditions:

- Editable team details presented.

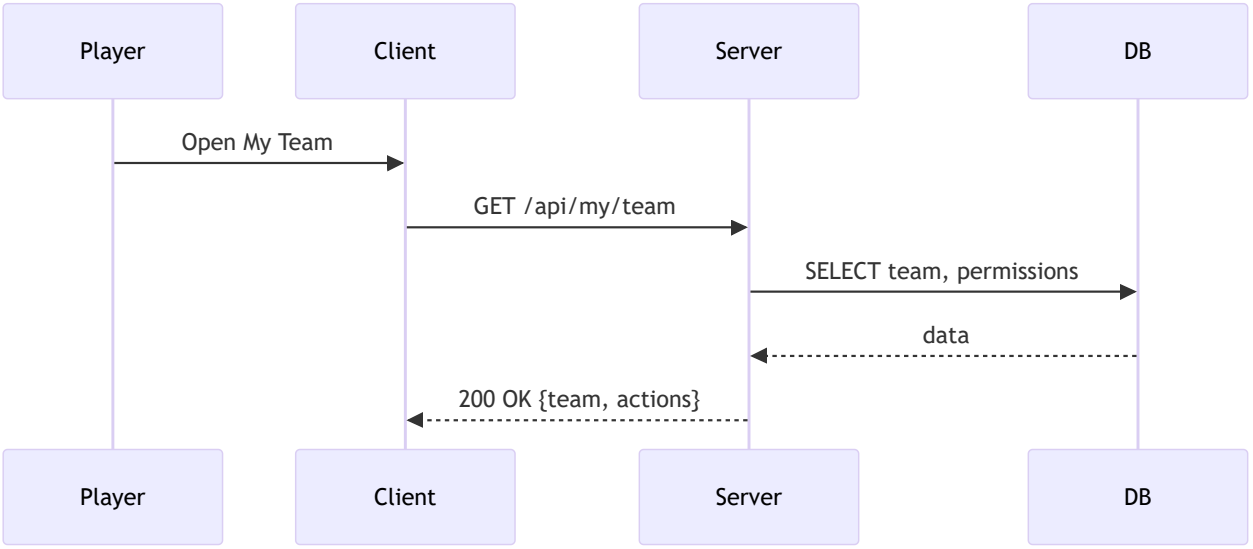
Normal Flow:

1. Client requests player's team data.
2. Server returns team, editable flags and available actions.

Acceptance Criteria:

- Team screen shows editable attributes and current budget.

Mermaid Diagram:



UC-14: Update Team Attributes

ID: UC-14

Title: Update Team Attributes

Description: Modify team attributes (budget allocations, HQ upgrades, staff assignments).

Actors:

- Player/Admin

Triggers:

- Player performs an update via UI.

Pre-conditions:

- Player authorized and has sufficient budget/resources.

Post-Conditions:

- Team attributes updated and persisted.

Normal Flow:

1. Client submits attribute changes.
2. Server validates business rules and budget.
3. Server updates DB and returns updated resource.

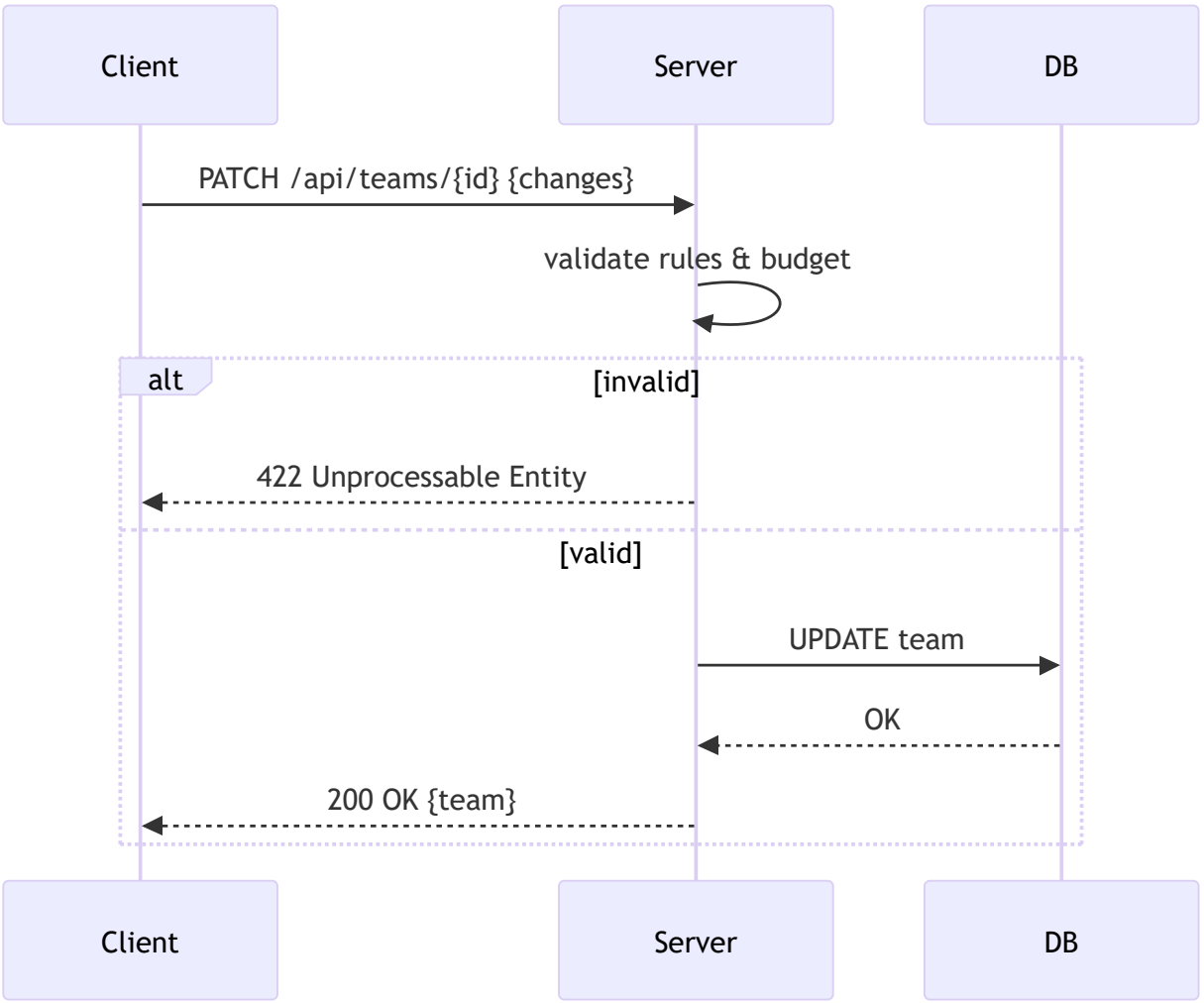
Alternative Flows:

- Insufficient funds: return 422 with required amount.

Acceptance Criteria:

- Updates persist and are reflected in subsequent reads.

Mermaid Diagram:



UC-15: Read Available Drivers

ID: UC-15

Title: Read Available Drivers

Description: Return list of uncontracted or transfer-available drivers for recruitment screens.

Actors:

- Player, Team AI

Triggers:

- User opens driver market or filter search.

Pre-conditions:

- Driver pool exists.

Post-Conditions:

- Filtered list returned.

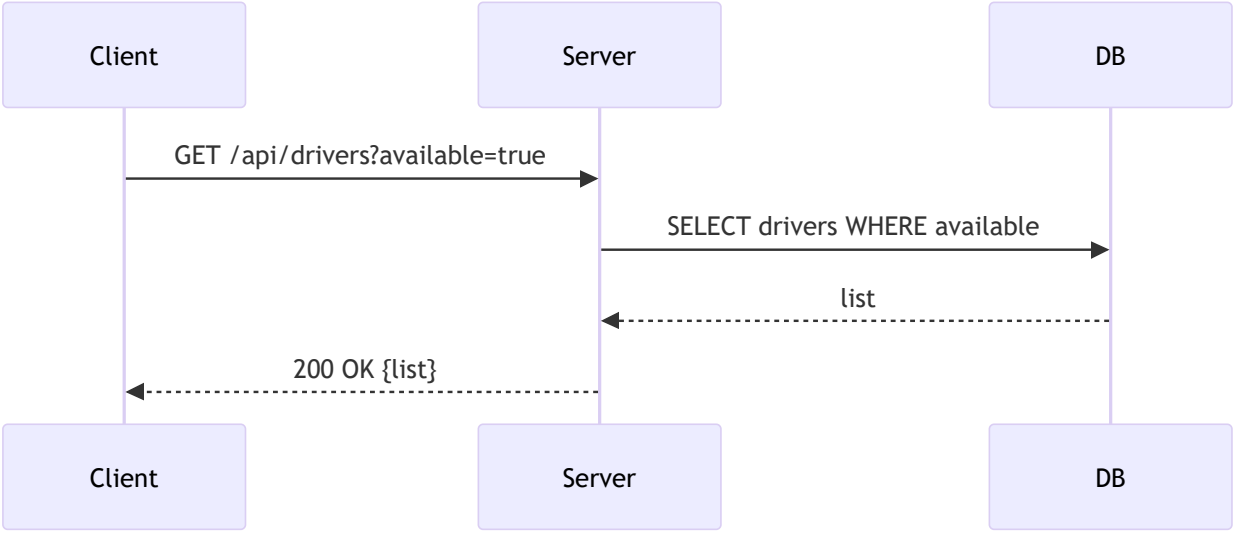
Normal Flow:

1. Client requests drivers with filters.
2. Server queries DB for available drivers and returns paginated results.

Acceptance Criteria:

- Response includes availability status and key attributes.

Mermaid Diagram:



UC-16: Update Driver Contract

ID: UC-16

Title: Update Driver Contract

Description: Sign, renew or terminate a driver contract and update team roster/budget.

Actors:

- Team Manager (player or AI), Admin

Triggers:

- Contract negotiation completion or admin action.

Pre-conditions:

- Driver and team exist; negotiation outcome determined.

Post-Conditions:

- Driver contract stored; team budget updated; availability toggled.

Normal Flow:

1. System receives contract terms for driver and team.
2. Server validates budget and contract rules.
3. Server saves contract and updates roster/budget.

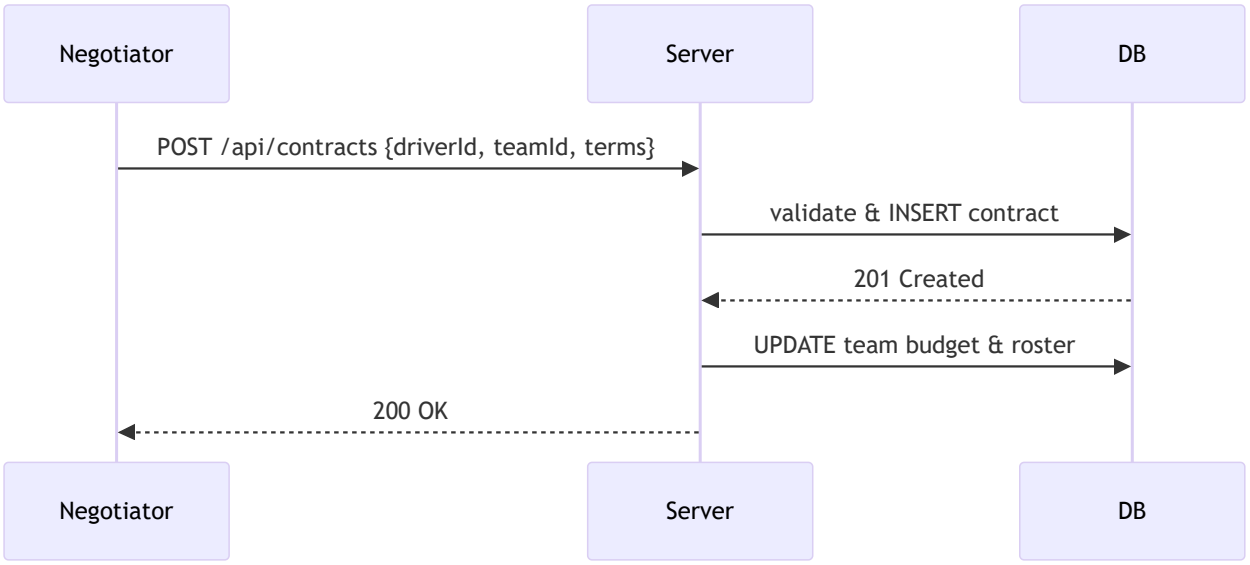
Alternative Flows:

- Budget shortfall: return 422.

Acceptance Criteria:

- Contract record exists and driver shows assigned team.

Mermaid Diagram:



UC-17: Read Available Staff

ID: UC-17

Title: Read Available Staff

Description: List staff available for hire with roles and modifiers.

Actors:

- Player, Team Manager

Triggers:

- Player opens staff hiring screen.

Pre-conditions:

- Staff pool exists.

Post-Conditions:

- List returned.

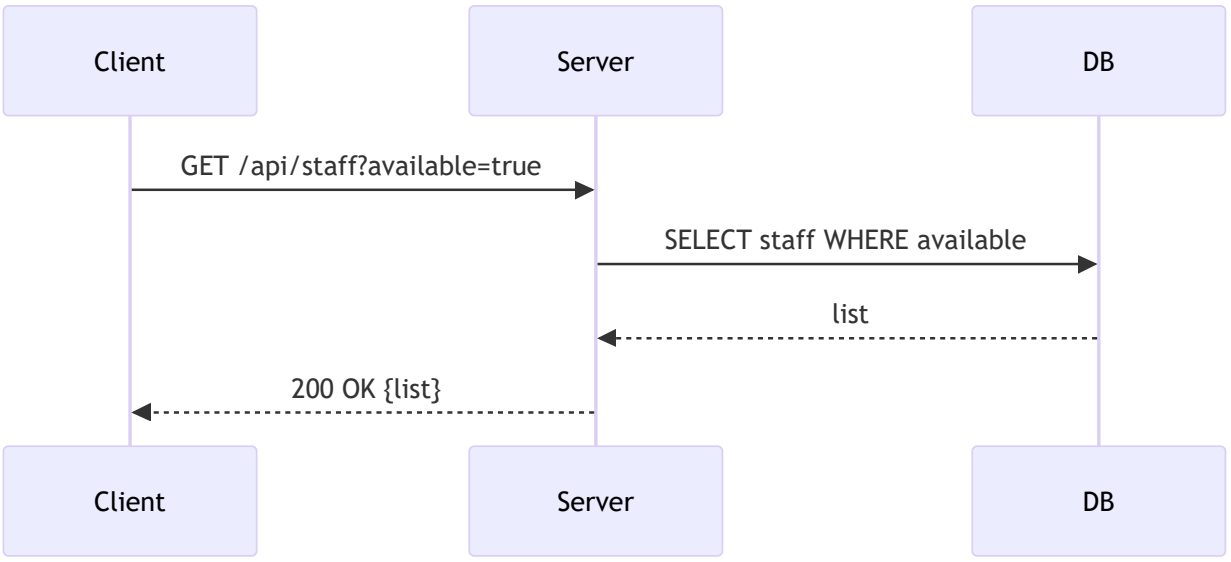
Normal Flow:

1. Client requests staff list with filters.
2. Server returns paginated pool with key stats and cost.

Acceptance Criteria:

- Includes role, cost, and performance modifiers.

Mermaid Diagram:



UC-18: Update Staff Contract

ID: UC-18

Title: Update Staff Contract

Description: Hire, fire, or renegotiate staff contracts and persist changes to team resources.

Actors:

- Player, Team Manager

Triggers:

- Hiring action completed.

Pre-conditions:

- Staff profile exists and is available.

Post-Conditions:

- Contract recorded; team budget and staff roster updated.

Normal Flow:

1. Client submits staff contract terms.
2. Server validates and stores contract; updates team state.

Acceptance Criteria:

- Staff assigned to team and salary deducted from budget.

Mermaid Diagram:

```
Parse error on line 3:
...ontract; UPDATE team    DB-->>Server: O
-----------------------^
Expecting 'SOLID_OPEN_ARROW', 'DOTTED_OPEN_ARROW', 'SOLID_ARROW',
'DOTTED_ARROW', 'SOLID_CROSS', 'DOTTED_CROSS', got 'NL'
```

UC-19: Update Driver Skills/Age (includes recycling retired drivers)

ID: UC-19

Title: Update Driver Skills/Age

Description: Periodic update of driver attributes (aging, skill progression/regression) and manage retirement/recycling.

Actors:

- Game Loop / Scheduler

Triggers:

- Season end or scheduled update tick.

Pre-conditions:

- Driver population present.

Post-Conditions:

- Driver attributes adjusted; retired drivers flagged and possibly recycled into pool.

Normal Flow:

1. Scheduler triggers attribute update job.
2. System applies progression/regression rules and increments ages.
3. Drivers surpassing retirement threshold are retired and optionally recycled.

Alternative Flows:

- Injury or special event triggers immediate attribute change.

Acceptance Criteria:

- Driver age and skills updated in DB; retired drivers moved to appropriate state.

Mermaid Diagram:

```
Parse error on line 5:
...kills; mark retirees    Server->>DB: UP
----------------------^
Expecting 'SOLID_OPEN_ARROW', 'DOTTED_OPEN_ARROW', 'SOLID_ARROW',
'DOTTED_ARROW', 'SOLID_CROSS', 'DOTTED_CROSS', got 'NL'
```

UC-20: Update Team Budget

ID: UC-20

Title: Update Team Budget

Description: Adjust team budget for income/expenses, sponsor payouts, race fees.

Actors:

- Finance System, Game Loop, Admin

Triggers:

- End of race, sponsor payment, purchase, or admin adjustment.

Pre-conditions:

- Team exists.

Post-Conditions:

- Team budget updated and persisted.

Normal Flow:

1. Transaction request to change team budget arrives.
2. Server validates transaction and updates budget atomically.
3. Server returns updated budget.

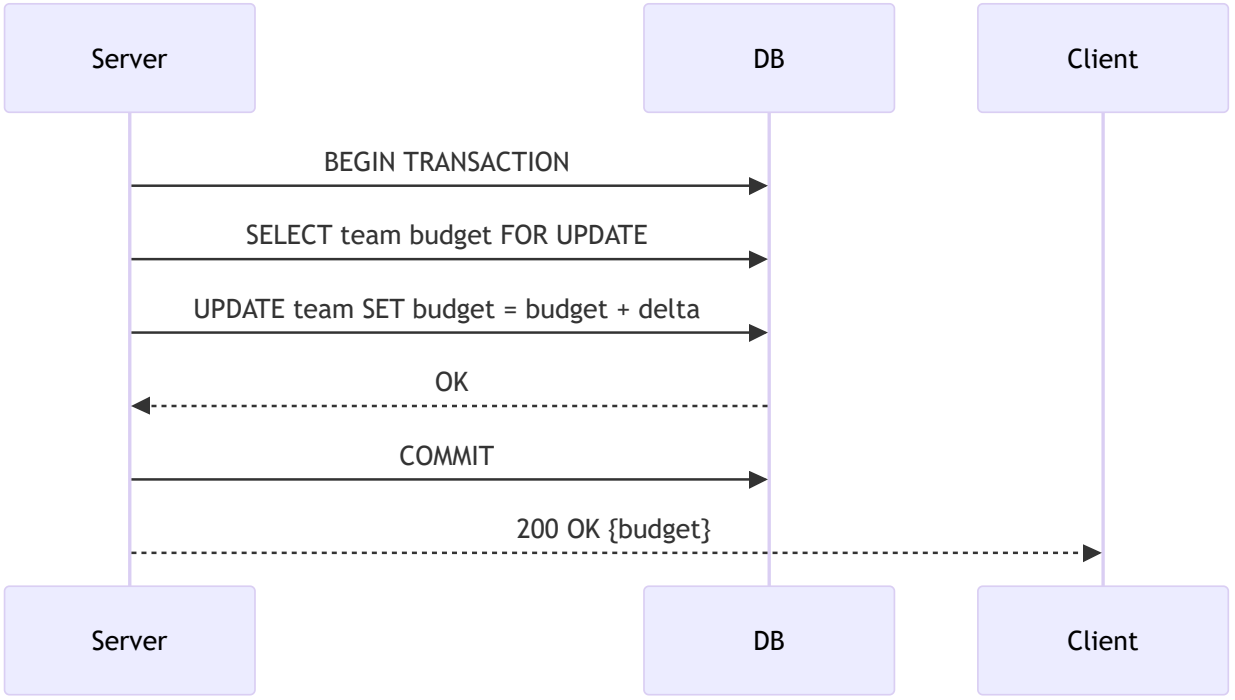
Alternative Flows:

- Insufficient funds for a debit: reject transaction.

Acceptance Criteria:

- Budget updates are atomic and consistent under concurrent access.

Mermaid Diagram:



LOW Priority (Polish & Advanced Features)

UC-21: Create New Driver (reuse pool)

ID: UC-21

Title: Create New Driver

Description: Create a new driver entry (usually by recycling or generating from faker) when pool needs expansion.

Actors:

- Admin, System

Triggers:

- Manual admin action or roster expansion event.

Pre-conditions:

- Driver pool may be exhausted or a new generation needed.

Post-Conditions:

- New driver available in pool.

Normal Flow:

1. Admin requests new driver creation.
2. Server generates or recycles a driver and inserts record.

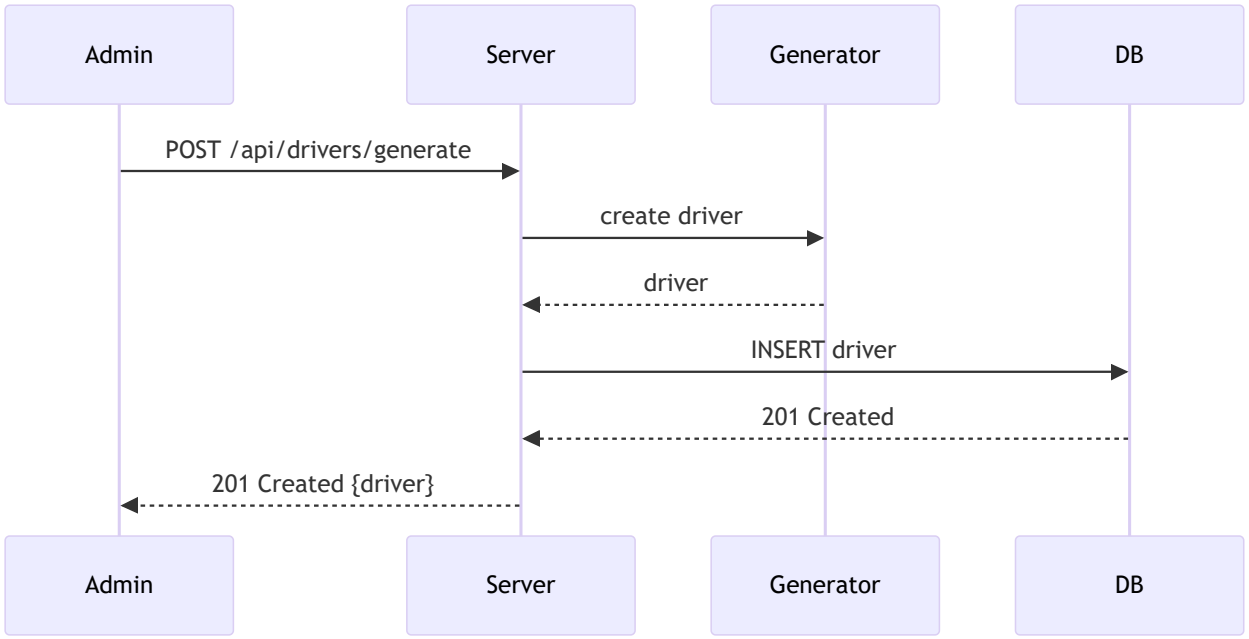
Alternative Flows:

- Duplicate name conflict: generate alternate.

Acceptance Criteria:

- New driver appears in available list with attributes.

Mermaid Diagram:



UC-22: Create New Staff Member (reuse pool)

ID: UC-22

Title: Create New Staff Member

Description: Create or recycle staff members for hire.

Actors:

- Admin, System

Triggers:

- Pool generation or admin action.

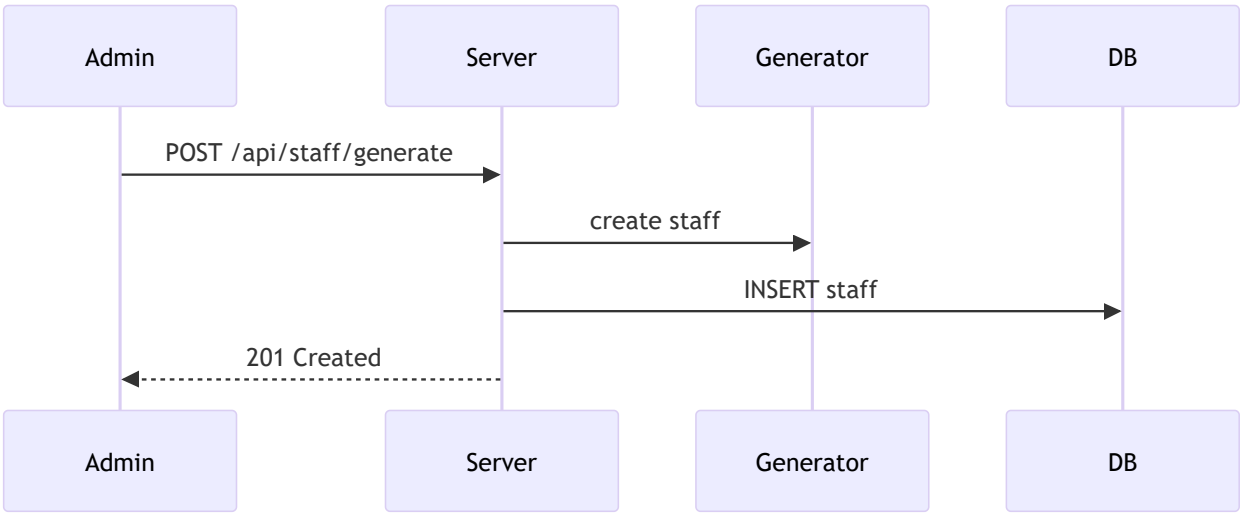
Post-Conditions:

- Staff record added.

Acceptance Criteria:

- Staff visible in available staff list.

Mermaid Diagram:



UC-23: Delete Championship

ID: UC-23

Title: Delete Championship

Description: Permanently remove a championship (admin only).

Actors:

- Admin

Triggers:

- Admin confirms deletion.

Pre-conditions:

- Admin permission and backup policy.

Post-Conditions:

- Championship removed, related resources archived or deleted per policy.

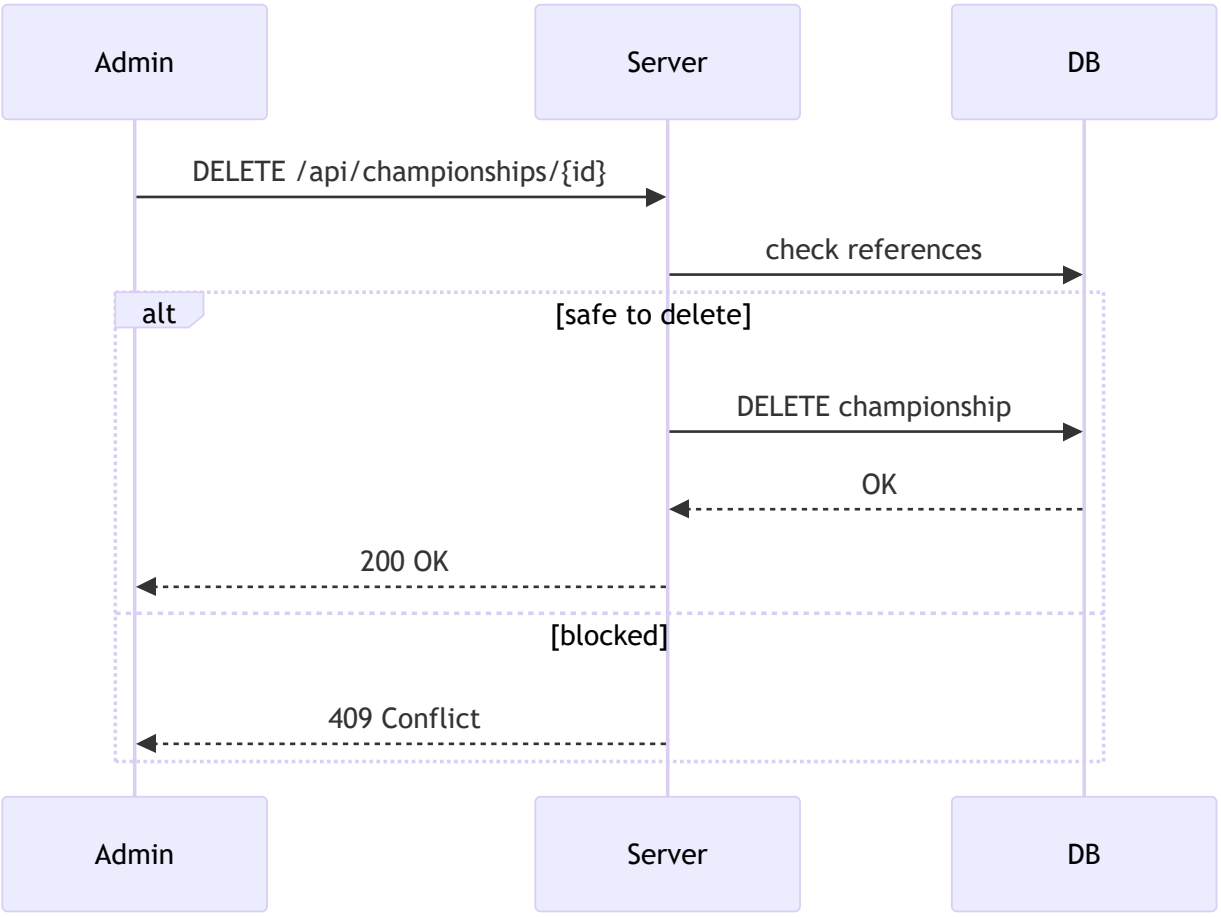
Normal Flow:

1. Admin requests deletion.
2. Server performs integrity checks and deletes or archives resources.

Exceptions:

- Referential integrity prevents deletion: return 409.

Mermaid Diagram:



UC-24: Create Custom Track

ID: UC-24

Title: Create Custom Track

Description: Add a custom track with characteristics for use in events.

Actors:

- Admin, Designer

Triggers:

- Track creation UI submission.

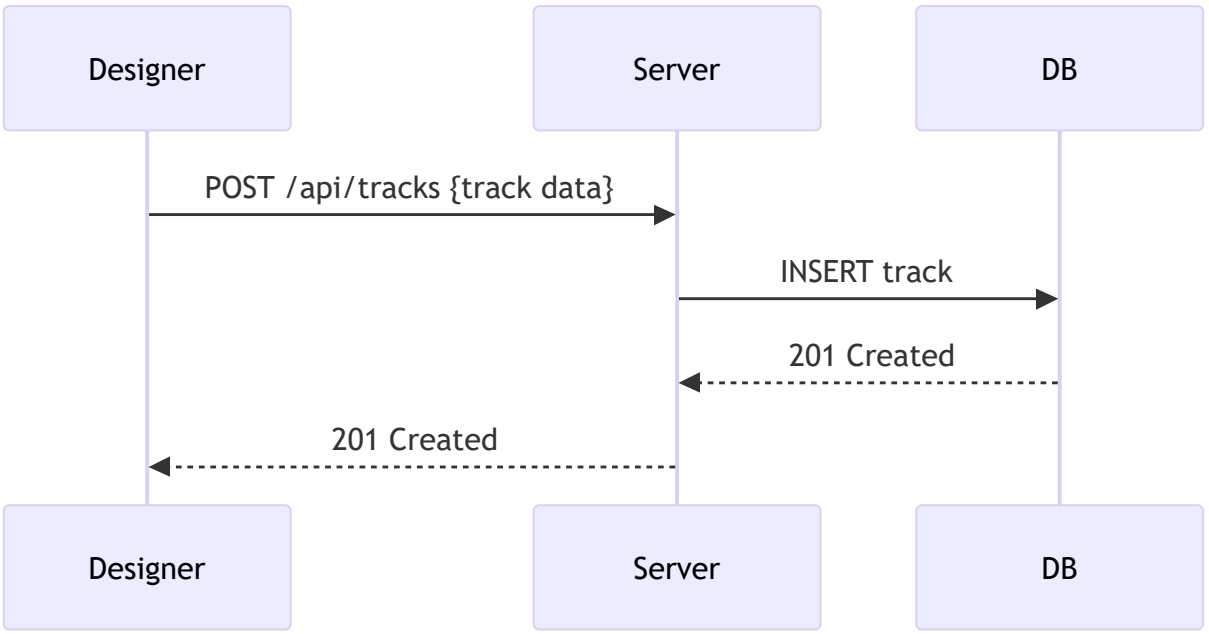
Post-Conditions:

- Track available for event creation.

Acceptance Criteria:

- Track metadata stored and usable in race config.

Mermaid Diagram:



UC-25: Update Track Characteristics

ID: UC-25

Title: Update Track Characteristics

Description: Modify track parameters (surface, length, grip) for balancing.

Actors:

- Admin

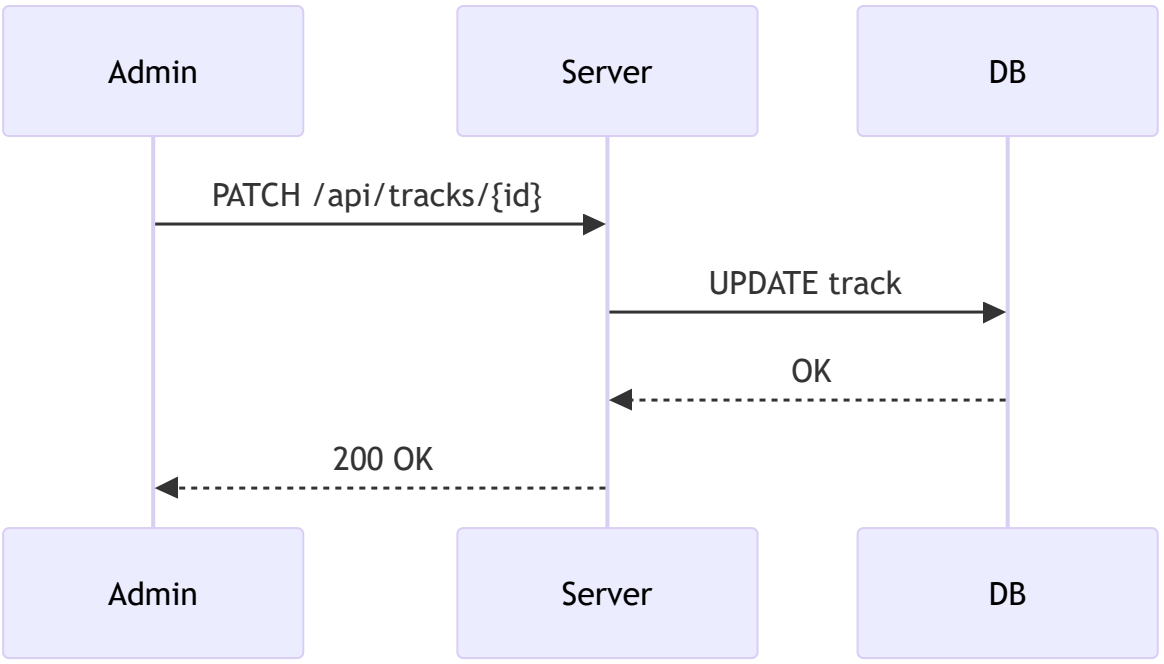
Triggers:

- Admin submits track updates.

Post-Conditions:

- Track updated and future events use new parameters.

Mermaid Diagram:



UC-26: Delete Track

ID: UC-26

Title: Delete Track

Description: Remove custom or unused tracks with checks for scheduled use.

Actors:

- Admin

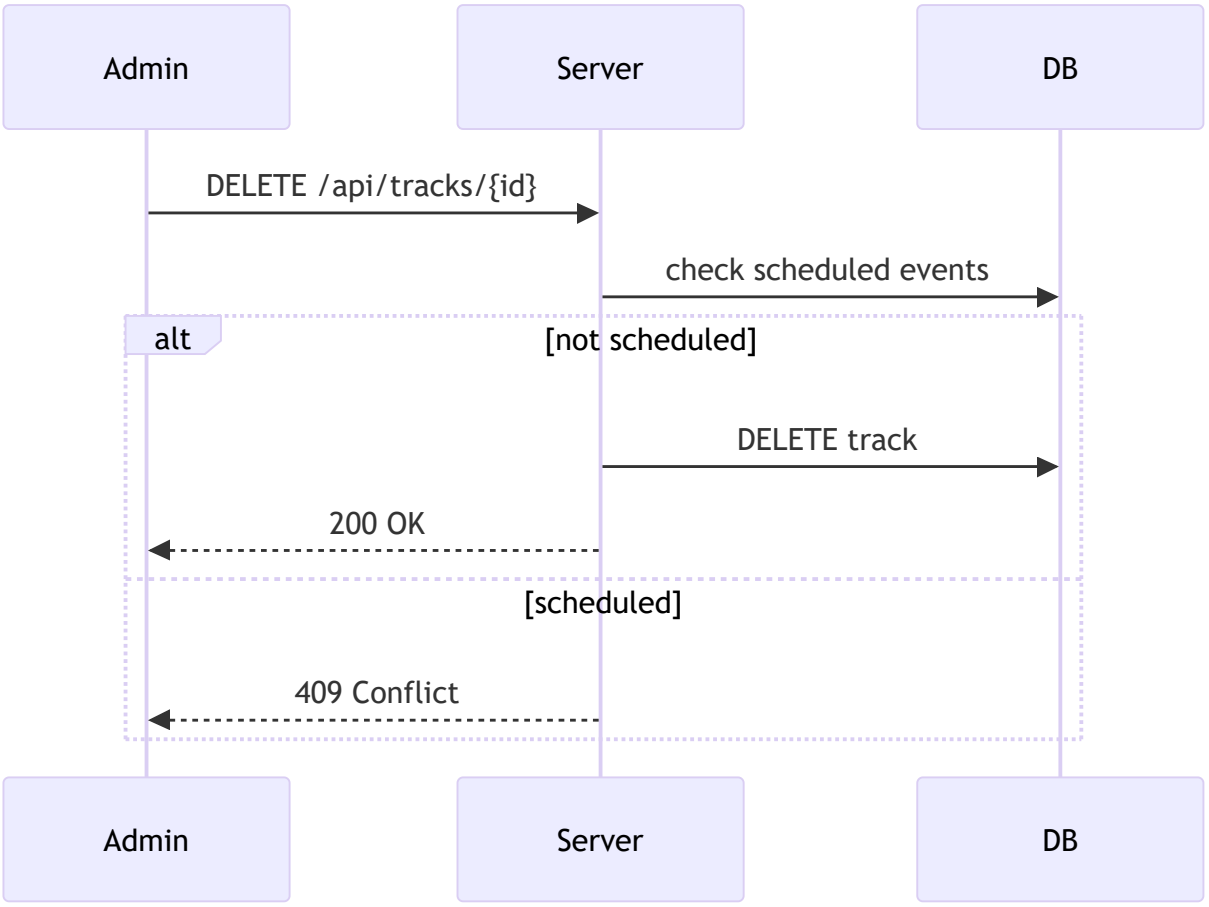
Triggers:

- Admin requests deletion.

Alternative Flows:

- Track scheduled in future: return 409 or force archive.

Mermaid Diagram:



UC-27: Read Staff Performance Modifiers

ID: UC-27

Title: Read Staff Performance Modifiers

Description: Return computed staff modifiers used by simulation (e.g., mechanic efficiency).

Actors:

- Simulation, UI

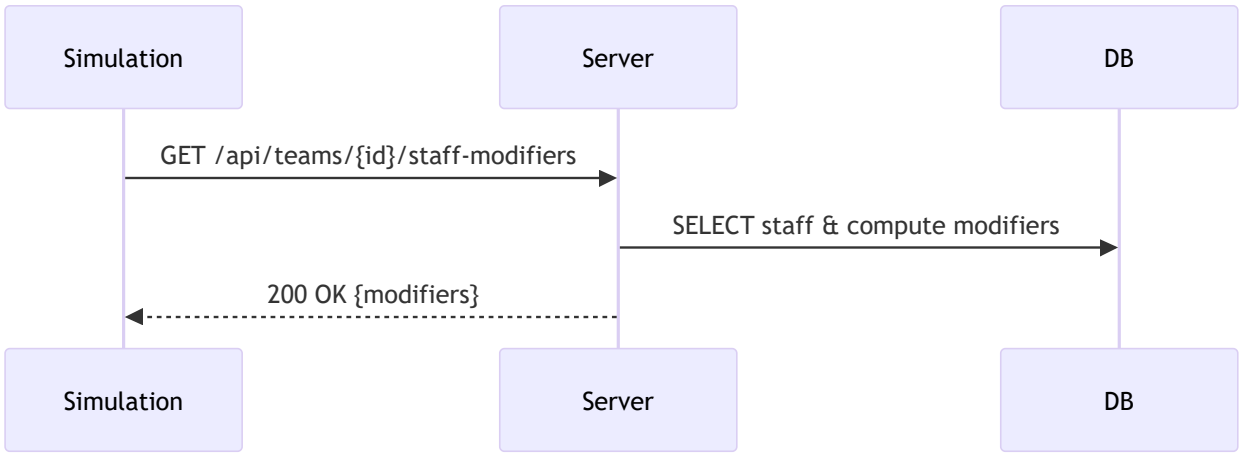
Triggers:

- Simulation requests modifier data for a team.

Post-Conditions:

- Modifiers returned.

Mermaid Diagram:



UC-28: Update Driver Appearance

ID: UC-28

Title: Update Driver Appearance

Description: Change cosmetic appearance for a driver (helmet, livery). Non-gameplay affecting.

Actors:

- Player

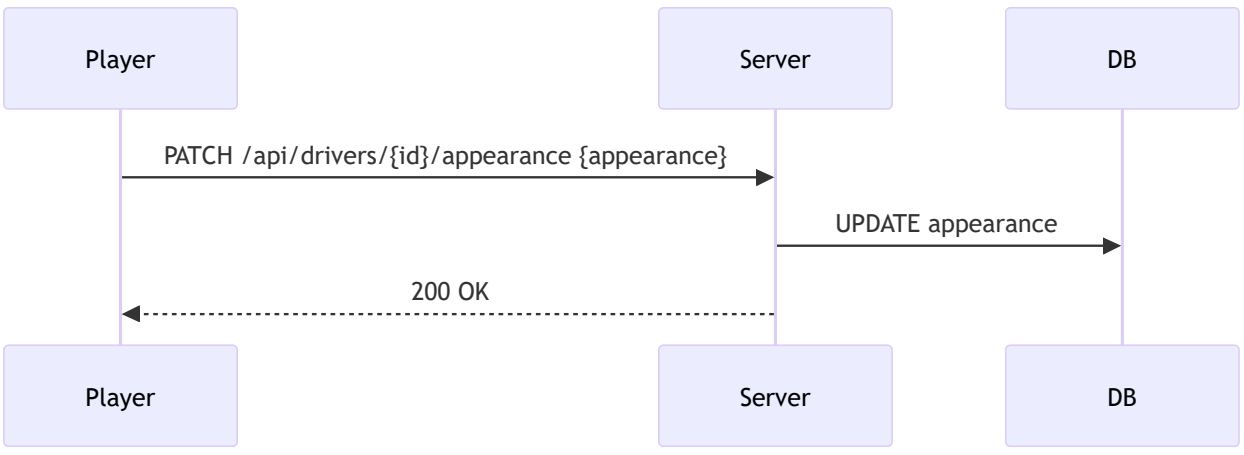
Triggers:

- Player saves appearance edits.

Post-Conditions:

- Appearance persisted and returned in reads.

Mermaid Diagram:



UC-29: Update Team Branding

ID: UC-29

Title: Update Team Branding

Description: Update team visual branding assets and metadata (logos, colors).

Actors:

- Player/Admin

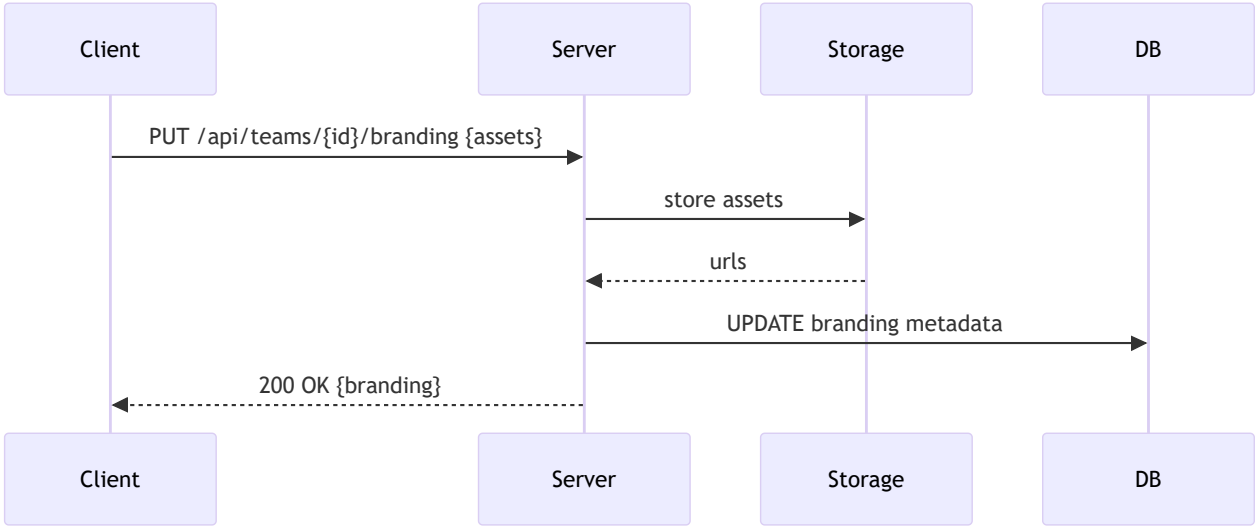
Triggers:

- Upload branding assets and save.

Post-Conditions:

- Branding stored and served to UI.

Mermaid Diagram:



End of use cases.